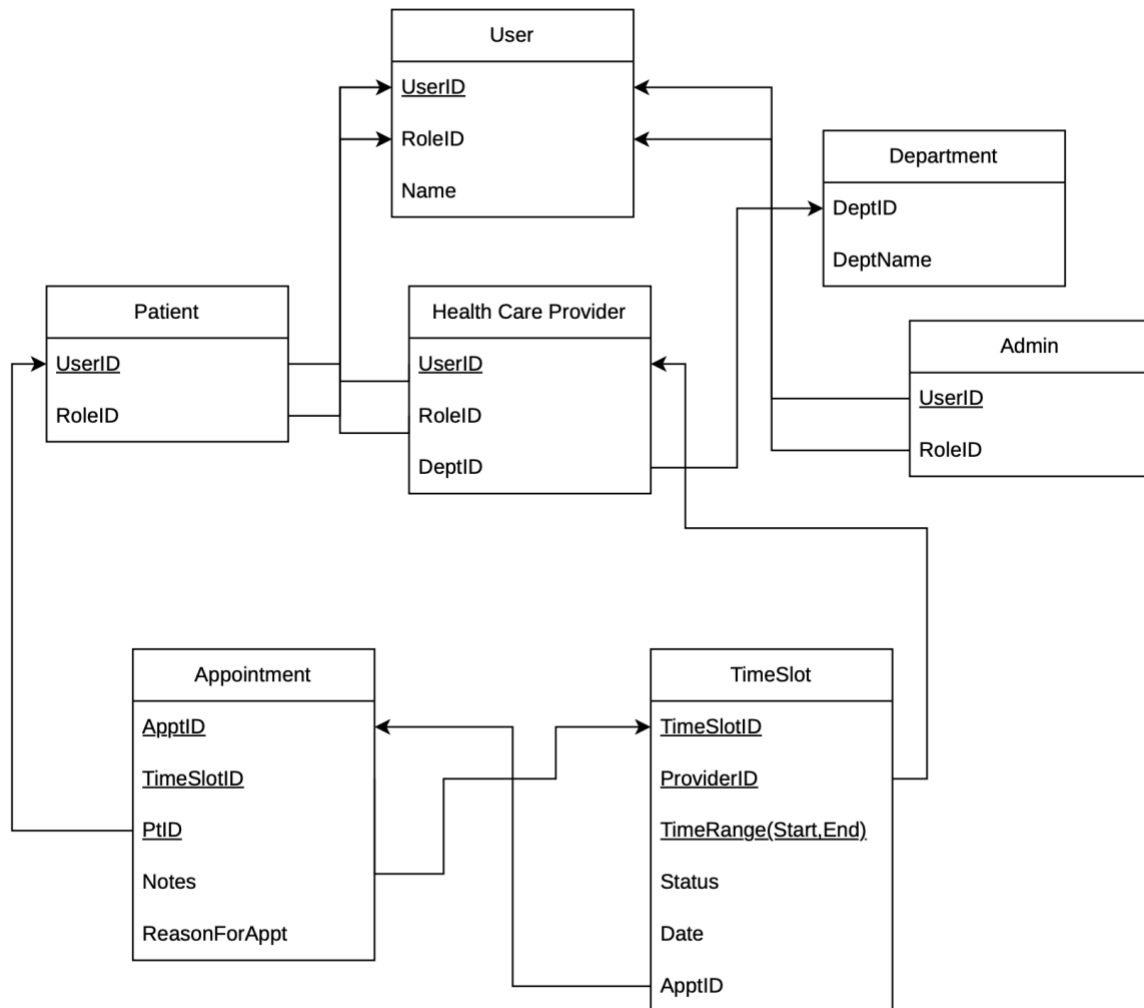


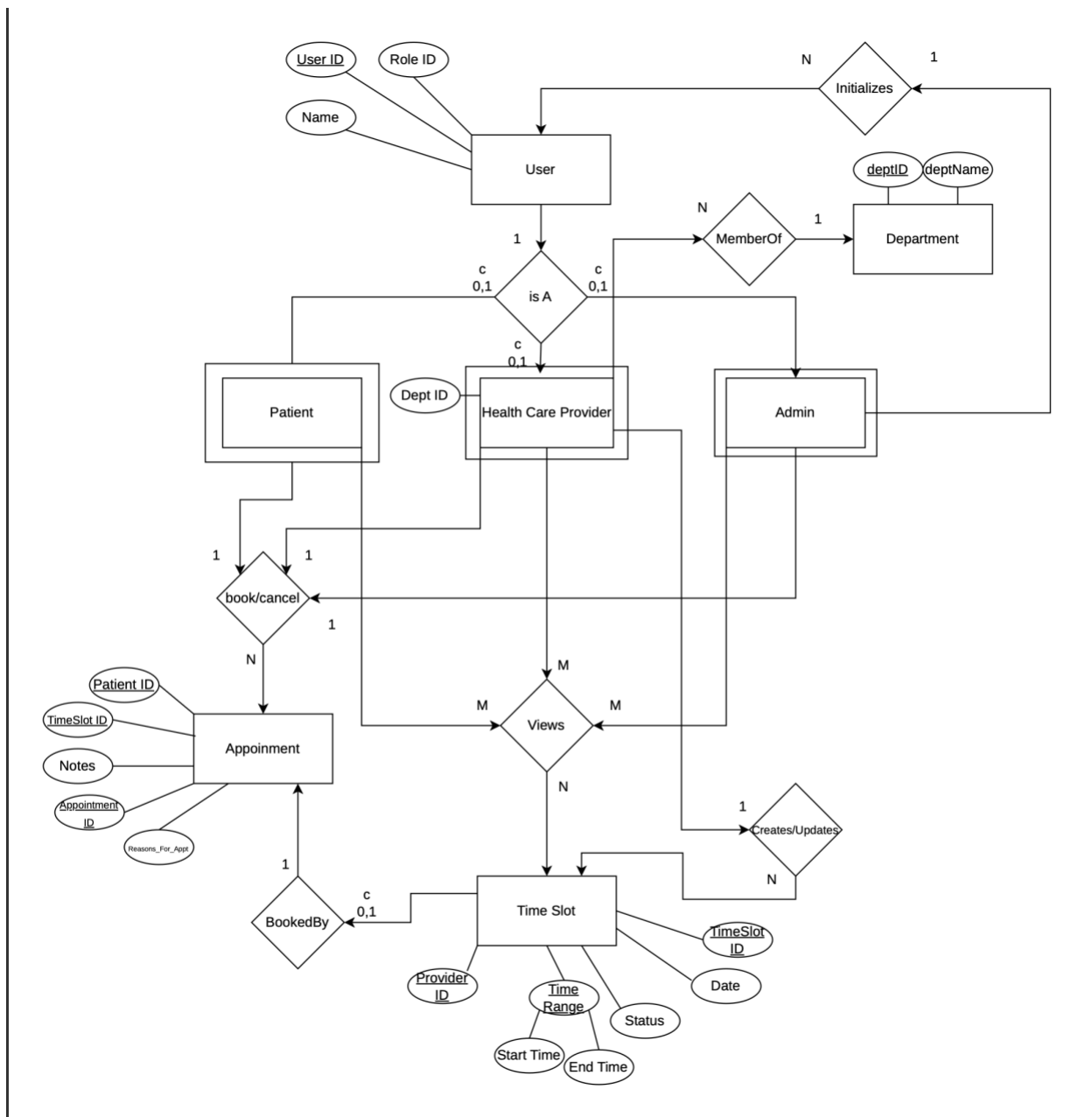
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CMPE 172

25 February 2026

Relational Schema and Explanation





Explanation:

I have 7 entities and 6 relationships.

The 7 entities are as follows: User, Patient, Health Care Provider, Admin, Department, Appointment, and Time Slot.

A User is one of the following Patient, Health Care Provider, or Admin thus the 1 – 0,1 relationship called “Is A”. A user must be one of those 3 roles. The RoleID indicates the Role. The 3 Roles are weak attributes because their Primary key of UserID is a foreign key of a User.

Health Care Provider has the foreign key of deptId which points to 1 Department. Each Department has a name and is allowed to have multiple Providers.

All user roles are allowed to book or cancel multiple appointments. Each Appointment must have 1 patient, 1 unique time slot, and an ID. These 3 attributes combine to form a unique Primary Key. Notes and Reason for appointment will change with each appt but can be non-unique.

Time Slots are very important. The primary key is composed of an ID, ProviderID, and Time Range which is a combination of a start and end Time. Time slots have a status (available/Booked) if booked they must have an appointment Id as foreign key for reference.

Each appointment must have an associated timeslotID which is a foreign key.

Each Patient is allowed to view all time slots in order to book appointments. Once an appointment has been booked the related time slot will change status and become unavailable to be booked. Patients may only view appointments they have booked.

Health Care Providers may create or update time slots so that new appointments can be booked. They are also allowed to view time slots and book appointments for patients.

Admin is allowed all view all time slots and may book appointments for any patients. They may not create time slots. Admin is also allowed to initialize new users assigning them roles and related privileges.